

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

- $\phi = 100$ at $x = 0$
- $\phi = 0$ at $x = L_x$
- Insulated boundaries ($\frac{\partial \phi}{\partial y} = 0$) at $y = 0$ and $y = L_y$

Project deliverables:

The main objectives of this project are as follows:

1. To formulate the two-dimensional steady-state diffusion equation using the Finite Difference Method.
2. To implement iterative numerical solvers (any 2 of your choice) for the discrete system or use PETSC where one must use multiple iterative solvers for performance comparison.
3. To apply and analyse the effects of different grid points.
4. To visualize the steady-state solution using contour and surface plots.